3.1

1.
The Grammar $S \to A, A \to \mathrm{a}^k\mathrm{b} \mid \mathrm{a}^k c$ is $\mathrm{LL}(k+1)$, because there are no words ux and uy with $(k+1) : x = (k+1) : y)$, but not $\mathrm{LL}(k)$, because $k : \mathrm{a}^k b = k : \mathrm{a}^k c$, but not $\beta = \gamma$.

2.
For every $\mathrm{LL}(k)$ grammar: $k : x = k : y \Rightarrow \beta = \gamma$. Also, $(k+1) : x = (k+1) : y \Rightarrow k : x = k : y$, so $(k+1) : x = (k+1) : y \Rightarrow \beta = \gamma$ also holds for the $\mathrm{LL}(k)$ grammar.

3.
An $\mathrm{LL}(0)$ grammar can have at most 1 word, as every production has to always produce the same result.

4.
The grammar has no non-productive terminals, so there are also productions for $A \underset{lm}{\overset{*}{\Longrightarrow}} w$ for some $w \in V_T^*$. Then:

$$S \underset{lm}{\overset{*}{\Longrightarrow}} uA\mu^k \underset{lm}{\Longrightarrow} uA\mu\mu^k \underset{lm}{\overset{*}{\Longrightarrow}} uw\mu^{k+1}$$
$$S \underset{lm}{\overset{*}{\Longrightarrow}} uA\mu^k \underset{lm}{\overset{*}{\Longrightarrow}} uw\mu^k$$

with $k : w\mu^{k+1} = w\mu^k$, but $A\mu \neq w$.